TEJAS MK2 FIGHTER JETS

A MEDIUM-WEIGHT CONTENDER

By Pradip R. Sagar

Barely a month after Chinese PLA fighter jets carried out sorties close to the friction points along the Line of Actual Control (LAC), the Cabinet Committee on Security (CCS) approved Rs 6,500 crore in additional funds to fast-track the development of an upgraded version of the indigenous Light Combat Aircraft (LCA) Tejas.

The Tejas Mk2 will use the more powerful GE-414 engines that will give the fighter jet more time in the air with heavier 'standoff weapon' loads. The Mk2 will be a medium-weight fighter (MWF) with an enlarged airframe and increased maximum take-off weight of 17,500 kg, compared to the 13,500 kg of the Mk1 version now. The Mk2 is being developed by the Aeronautical Development Agency (ADA), which says the critical design and drawings are ready. The first prototype design was to roll out in August 2022, but the new cut-off date is the year-end. ADA says the first prototype will take flight by 2024 and production will start by 2027. Hindustan Aeronautics Ltd (HAL) will manufacture the jets for IAF. If all goes to plan, the Mk2 will replace 16 of the IAF's ageing fighter jet squadrons, and complement the Sukhoi and Rafale jets in the future.

In 2008, when the CCS initially approved the project, the LCA Mk2 was supposed to be in the lightweight category. Subsequently, its developers and the IAF went on adding requirements and, eventually, by 2017, it had become a medium-weight fighter. The initial sanctioned budget of Rs 2,800 crore soon became inadequate and fresh funds of Rs 6,500 crore was injected to fabricate four prototypes and onboard electronics systems development like avionics, radar and an early warning unit. After weaponisation, flight testing and new systems are finalised, sources say the indigenous content of Mk2 could be around 75 per cent, unlike Tejas LCA Mk1, where it is less than 60 per cent. Air Chief Marshal Vivek Ram Chaudhary, while ruing his force's depleted jet squadron strength, says the IAF is betting big on the Mk2 to deliver, and on time.

Besides Tejas, the Centre is also contemplating green-lighting the fifth generation Advanced Medium Combat Aircraft (AMCA) programme with an estimated budget of Rs 15,000 crore. The AMCA will reportedly have stealth and supercruise features and could put India in a select group of countries (US, Russia and China are the others) that have such capabilities.

The IAF has placed a Rs 48,000 crore order for 83 Mk-1A variants with HAL, and is also hoping to induct six squadrons of Mk2 in the future. HAL says it is targeting 2024 for delivery of the first Mk-1A jets and completing the order book by 2029. ■

17,500* KG
Maximum take-off weight

2,385 KMPH
Max. speed (Mach 1.8)

6,500 KG
Payload capacity

₹6,500 CRORE
to develop the first four prototype jets

2,500 KM
Combat range

75%
Indigenous content of the medium-weight fighter



F414-GE-INS6 ENGINE
The 4.5 Gen LCA-Mk2's superior engine will allow more air time and heavier standoff weapons payloads—Spice-2000 bombs and Scalp, Crystal Maze and BrahMos missiles—for deep penetration offensive missions

* The numbers are all proposed figures

3D artwork of LCA Tejas Mk-2 by KUNTAL BISWAS